AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Type 1: Research Project

Explainable Bimodal Deepfake Detection for Image and Video Analysis

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Feedback Integration (Panel Review 1)

- **Research Alignment:** Switch to Type 1 (Research-based) project.
- **Project Focus:** Removed "Forensic" focus to shift alignment towards deepfake detection for social media use case.
- **Consulting the Law Department regarding real world use cases:** Irrelevant to the project focus due to dropping the "Forensic" focus
- **Courtroom Datasets:** Irrelevant to the project focus due to dropping the "Forensic" focus

Outline

Presentation Overview

01 Demonstration of Gaps

02 Demonstration of Need

03 Challenges

04 Problem Statement

05 Originality and Novelty

06 Objectives and Deliverables

07 Review of Study

08 Proposed Methodology

09 Proposed Architecture

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1. Demonstration of Gaps

Hypothesis Validation & Problem Formulation

- From our literature survey, we found no end-to-end deepfake detection system that jointly combines:
  - **Adversarial Robustness**: Existing architectures often fail when subjected to adversarial manipulations and noise.
  - **Compression Preprocessing**: Existing architectures often fail when subjected to heavy compression artifacts that are common on social media.
  - **Bimodality**: Most current models focus exclusively on spatial features (images) or temporal inconsistencies (videos), rarely both, in a combined framework.
  - **Explainability**: High-performing CNN/Transformer models lack Explainability (XAI), making it impossible for users to know why a frame was flagged.
  - **Cross-Dataset Generalization:** Significant performance drop-off when models trained on one GAN-type are tested on newer diffusion-based deepfakes.
- This gap motivates the need for an explainable, robust, and modal deepfake detection framework designed specifically for image and video analysis

- **Misinformation:** Generated deepfakes could be easily used to spread misinformation [7].
- **Defamation of Public Figures:** There have been numerous instances of face swap videos of celebrities created to harm their reputation and brand [7].
- **Human Limitation:** Humans are becoming incapable of manually distinguishing digitally altered content [16].

3. Challenges

Hypothesis Validation & Problem Formulation

- Most existing deepfake detection systems function as black-box models, providing only confidence scores without interpretable explanations.
- Explainability, when present, is often not evaluated in deepfake detection architectures.
- Current detection models often fail to reliably identify deepfakes once the media has been degraded by standard social media compression and visual distortions
- Many existing detection methods fail to handle unseen attacks in open set scenarios, struggling to identify deepfakes generated by different schemes or algorithms than those used during training.
- Existing detection algorithms require substantial computation and are not designed to operate under real-time constraints, which is critical given the rapid spread of content on social media.

Redefined Problem Statement

Deepfakes are becoming increasingly convincing, and while AI detectors can identify them with reasonable accuracy, they rarely explain why something is flagged. That's a significant limitation when real world decisions depend on the outcome. Our proposed system goes beyond binary detection by showing its reasoning, highlighting the specific facial regions or video frames that reveal manipulation, through visual heatmaps and frame by frame timelines. Rather than a black box classifier, it functions as a transparent tool giving media analysts the clarity and evidence needed to make confident, well informed judgments.

5. Originality and Novelty

Rubric: Hypothesis Validation & Problem Formulation

- **Problem Context:** Current implementations of Deepfake Detection Systems are done primarily with the help of black-box models like CNNs, providing limited interpretability into the decisions made by the model.
- **Innovation Type:**
  - **Adaptation:** Applying Explainable Deepfake Detection .
  - **Improvement:** Enhancing existing Deepfake Detection models by
    - Incorporating a hybrid approach to allow for both image and video processing
    - Implementing explainability to provide real-world usability
    - Performing Adversarial Robustness Training(ART) to improve resilience to adversarial attacks.

6. Objectives & Deliverables

Rubric: Hypothesis Validation & Problem Formulation

Objective	Deliverable
1: Proposing the Solution	Defining problem statement, system architecture design, selection of models , and identification of datasets.
2: Implementation	Developing deepfake detection system including preprocessing pipeline, trained models, and integrating image and video detection modules.
3: Comparison with SOTA Models	Performance evaluation comparing proposed system with existing models using accuracy and processing time metrics.
4: Improvement and Optimization	Enhancing model through optimization techniques, improving robustness and efficiency, and refining system performance .

7. Review of Study (1/3)

Rubric: Study Methodology

Title	Authors	Journal - Year	Description
Multimodal Fake News Detection on Social Media Using Deep Learning	Maged Nasser, Noreen Izza Arshad, Abdulalem Ali, Hitham Alhussian, Faisal Saeed, Aminu Da'u, Ibtehal Nafea	Results in Engineering - 2025	This paper focuses on detecting fake news and deepfake content on social media using multimodal deep learning. It combines textual features (using models like BERT), visual features (using CNN architectures), and in some cases audio features. The methodology involves extracting features from multiple modalities, followed by applying fusion techniques (early fusion and late fusion) and performing classification using deep neural networks. The models improve detection accuracy by capturing inconsistencies between textual and media content. The study also highlights the importance of incorporating Explainable AI (XAI) to enhance transparency and build user trust in deepfake detection systems.
Combating Identity Forgery in Social Media: Detecting Manipulated Profile Videos With a Motion-Texture Fusion Framework	Yanan Song, Jiaming Kuang and Ronghua Xu	IEEE Access - 2026	This paper addresses the problem of identity forgery and deepfake videos circulating on social media platforms. With the advancement of AI-based generative models, attackers can create highly realistic fake profile videos that can be used for impersonation, misinformation, and fraud. To solve this problem, the authors propose a deep learning framework called SFE-Net (Spatio-temporal Fusion Extraction Network) for detecting manipulated profile videos.
Domain-Adaptive Representation Learning for Multimodal Deepfake Detection	Ijaz Ul Haq and Khalid M. Malik	IEEE Transactions on Computational Social Systems - 2025	This paper addresses the increasing threat posed by multimodal deepfakes on social media, where manipulated content combines both visual and audio elements to create highly realistic forgeries. Traditional detection methods, which often rely on a single modality, struggle to identify such sophisticated manipulations and fail to generalize across different datasets and attack types. To tackle this issue, the authors propose a novel framework called DARL, which treats deepfake detection as an audio visual synchronization problem. The core idea is that deepfakes disrupt the natural alignment between lip movements and speech, making inconsistencies in LipSync a reliable indicator of forgery.

7. Review of Study (2/3)

Rubric: Study Methodology

Title	Authors	Journal - Year	Description
Securing online integrity: a hybrid approach to deepfake detection and removal using Explainable AI and Adversarial Robustness Training	R. Uma Maheshwari and B. Paulchamy	Automatika - 2024	The above paper discusses a unique approach to deepfake detection for visual inputs. The paper utilizes Convolutional Neural Network(CNN) augmented with Adversarial Robustness Training(ART). Adversarial attacks take advantage of the vulnerability of deep learning models that causes huge change in output with minor perturbations to the input. ART is a method of improving a model's performance against adversarial attacks. It does this by creating an augmented dataset containing both clean images and images that have been imperceptibly tampered. The model is then trained on this dataset forcing it to learn features that are immune to perturbations. The paper also integrated the model after training with the Local Interpretable Model-agnostic Explanations(LIME) and SHapley Additive exPlanations(SHAP) to provide visual reasoning for why the model classified the input into a certain class.
DeepExplain: Enhancing DeepFake Detection Through Transparent and Explainable AI model	Venkateswarlu Sunkari, A. Srinagesh	Informatica - 2024	The above paper proposes an explainable method of deepfake detection titled 'DeepExplain' that utilizes CNNs with Long Short-Term Memory(LSTM) to detect anomalies in a video. The LSTM allows the CNN capture and retain context from previous frames allowing it to detect anomalies in videos. DeepExplain also makes use Grad-CAM and SHAP values for providing explainability to the results obtained. It uses the DFDC dataset for training and testing. It also compared CNN, RNN, LSTM and GAN, but CNN with LSTM provided the highest accuracy.

7. Review of Study (3/3)

Rubric: Study Methodology

Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence	Sajid Ali, Tamer Abuhmed, Shaker El-Sappagh, Khan Muhammad, Jose M. Alonso-Moral, Roberto Confalonieri, Riccardo Guidotti, Javier Del Ser, Natalia Díaz-Rodríguez, Francisco Herrera	Information Fusion - 2023	The above study provides an overview of the current research and trends in the field of eXplainable AI(XAI). It provides a background on XAI, common definitions and current research directions. It divides XAI techniques into four categories: · Data Explainability: To foster a better understanding of the datasets used to train models. · Model Explainability: For understanding the inner workings of models to make more optimized models in the future. · Post-hoc Explainability: Focuses on providing reasoning for explaining the decision taken by a model w/ respect to a particular input. · Assessment of Explainability: To evaluate the effectiveness of XAI techniques i.e, how well an XAI technique explains a model and how accurate it is. The paper also provides recommendations for software tools for XAI as well as future research directions.

8. Proposed Methodology

Components:

- **Input:** User-submitted images (RGB,shape:$(H, W, 3)$) and videos (after frame extraction:$(T, H, W, 3)$).
- **Preprocessing Technique:** Compression Based Preprocessing
- **Training Enhancement:**
  - **Adversarial Robustness Training**
  - **Data Augmentation**
- **Models Used:**
  - **XceptionNet:** For image-based deepfake detection.
  - **EfficientNet + LSTM:** For video-based detection .
- **Explainability Module:** GRAD-CAM for generating visual heatmaps.
- **Output:** Real/Fake label with confidence score + visual explanation heatmap.

System Architecture Diagram

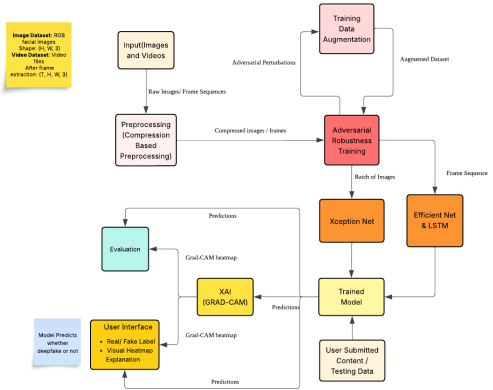


Figure 1: Architecture Diagram for Social Media Deepfake Detection

10. Distribution of work along phases

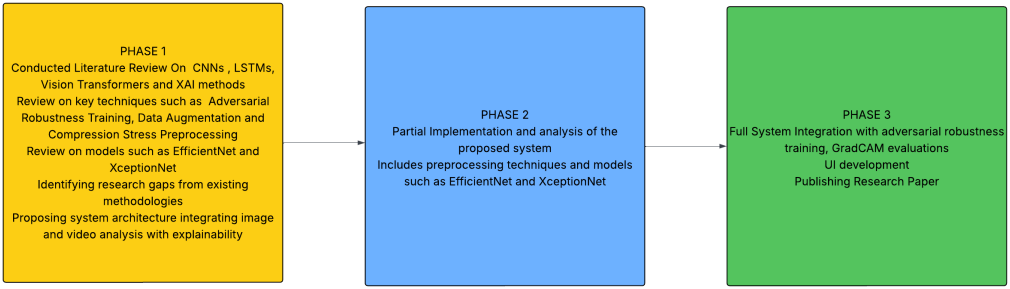


Figure 2: Proposed Timeline

References I

[1] M. S. Momin, A. Sufian, D. Barman, M. Leo, C. Distante, and N. Damer, “Explainable deepfake detection across different modalities: An overview of methods and challenges,” *Image and Vision Computing*, pp. 105 738–105 738, 09 2025.

[2] A. Almestekawy, H. H. Zayed, and A. Taha, “Deepfake detection: Enhancing performance with spatiotemporal texture and deep learning feature fusion,” *Egyptian Informatics Journal*, vol. 27, p. 100535, 09 2024.

[3] S. Mathews, S. Trivedi, A. House, S. Povolny, and C. Fralick, “An explainable deepfake detection framework on a novel unconstrained dataset,” *Complex & Intelligent Systems*, vol. 9, 01 2023.

[4] N. Mansoor and A. I. Iliev, “Explainable ai for deepfake detection,” *Applied Sciences*, vol. 15, pp. 725–725, 01 2025. [Online]. Available: https://www.mdpi.com/2076-3417/15/2/725

References II

[5] M. M. Alwateer, “Explainable deep fake framework for images creation and classification,” *Journal of Computer and Communications*, vol. 12, pp. 86–101, 2024.

[6] L. H. Singh, P. Charanarur, and N. K. Chaudhary, “Advancements in detecting deepfakes: Ai algorithms and future prospects - a review,” *Discover Internet of Things*, vol. 5, 05 2025.

[7] M. Masood, M. Nawaz, K. M. Malik, A. Javed, A. Irtaza, and H. Malik, “Deepfakes generation and detection: State-of-the-art, open challenges, countermeasures, and way forward,” *Applied Intelligence*, vol. 53, 06 2022. [Online]. Available: https://link.springer.com/article/10.1007/s10489-022-03766-z

[8] R. U. Maheshwari and B. Paulchamy, “Securing online integrity: a hybrid approach to deepfake detection and removal using explainable ai and adversarial robustness training,” *Automatika*, vol. 65, pp. 1517–1532, 09 2024. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/00051144.2024.2400640

References III

[9] S. Venkateswarulu and A. Srinagesh, “Deepexplain: Enhancing deepfake detection through transparent and explainable ai model,” *Informatica*, vol. 48, 05 2024. [Online]. Available: https://www.informatica.si/index.php/informatica/article/view/5792

[10] B. Gowrisankar and V. L. Thing, “An adversarial attack approach for explainable ai evaluation on deepfake detection models,” *Computers & security*, vol. 139, pp. 103 684–103 684, 04 2024.

[11] S. Ali, T. Abuhmed, S. El-Sappagh, K. Muhammad, J. M. Alonso-Moral, R. Confalonieri, R. Guidotti, J. D. Ser, N. Díaz-Rodríguez, and F. Herrera, “Explainable artificial intelligence (xai): What we know and what is left to attain trustworthy artificial intelligence,” *Information Fusion*, vol. 99, p. 101805, 04 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253523001148

References IV

[12] F. Khalid, A. Javed, K. M. Malik, and A. Irtaza, “Explanet: A descriptive framework for detecting deepfakes with interpretable prototypes,” *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 6, pp. 486–497, 10 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10542403

[13] R. Rafique, R. Gantassi, R. Amin, J. Frnda, A. Mustapha, and A. H. Alshehri, “Deep fake detection and classification using error-level analysis and deep learning,” *Scientific Reports*, vol. 13, p. 7422, 05 2023. [Online]. Available: https://www.nature.com/articles/s41598-023-34629-3

[14] S. A. Khan and D.-T. Dang-Nguyen, “Deepfake detection: Analysing model generalisation across architectures, datasets and pre-training paradigms,” *IEEE Access*, vol. 12, pp. 1880–1908, 01 2024.

References V

[15] G. Gupta, K. Raja, M. Gupta, T. Jan, S. T. Whiteside, and M. Prasad, “A comprehensive review of deepfake detection using advanced machine learning and fusion methods,” *Electronics*, vol. 13, pp. 95–95, 12 2023.

[16] I. Balafrej and M. Dahmane, “Enhancing practicality and efficiency of deepfake detection,” *Scientific Reports*, vol. 14, 12 2024.

[17] A. H. Soudy, O. Sayed, H. Tag-Elser, R. Ragab, S. Mohsen, T. Mostafa, A. A. Abohany, and S. O. Slim, “Deepfake detection using convolutional vision transformers and convolutional neural networks,” *Neural Computing and Applications*, vol. 36, 08 2024.

[18] T. Qiao, S. Xie, Y. Chen, F. Retraint, R. Shi, and X. Luo, “Deepfake detection fighting against noisy label attack,” *IEEE Transactions on Multimedia*, vol. 26, pp. 9047–9059, 01 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10491334

References VI

[19] Y. Xu, P. Terhörst, M. Pedersen, and K. Raja, “Analyzing fairness in deepfake detection with massively annotated databases,” *IEEE Transactions on Technology and Society*, vol. 5, pp. 1–1, 2024. [Online]. Available: https://arxiv.org/pdf/2208.05845.pdf#:~:text=The%20Deepfake%20detection%20backbone%20methods

[20] K. N. Ramadhani, R. Munir, and N. P. Utama, “Improving video vision transformer for deepfake video detection using facial landmark, depthwise separable convolution and self attention,” *IEEE Access*, vol. 12, pp. 8932–8939, 01 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10388363

[21] L. A. Motalib, O. Mustapha, and H. Mustapha, “Compression-aware hybrid framework for deep fake detection in low-quality video,” *IEEE Access*, vol. 13, pp. 131 980–131 997, 2025.

References VII

[22] D. Xiong, Z. Wen, C. Zhang, D. Ren, and W. Li, “Bmnet: Enhancing deepfake detection through bilstm and multi-head self-attention mechanism,” *IEEE Access*, vol. PP, p. 1, 01 2025. [Online]. Available: https://www.researchgate.net/publication/388379963_BMNet_Enhancing_Deepfake_Detection_through_BiLSTM_and_Multi-Head_Self-Attention_Mechanism

[23] X. Liu, Y. Yu, X. Li, and Y. Zhao, “Mcl: Multimodal contrastive learning for deepfake detection,” *IEEE transactions on circuits and systems for video technology*, vol. 34, pp. 1–1, 01 2023.

[24] X. Jin, Y. Kou, Y. Xie, Y. Zhao, M. Laiha, Q. Jiang, and W. Zhou, “Learning local texture and global frequency clues for face forgery detection,” *Biomimetics*, vol. 10, pp. 480–480, 07 2025.

References VIII

[25] L. Cirillo, A. Gervasio, and I. Amerini, “Explainability-driven adversarial robustness assessment for generalized deepfake detectors,” *EURASIP Journal on Information Security*, vol. 2025, 08 2025.

[26] P. Sharma, M. Kumar, and H. K. Sharma, “A robust ensemble model for deepfake detection of gan-generated images on social media,” *Discover Computing*, vol. 28, 04 2025. [Online]. Available: https://link.springer.com/article/10.1007/s10791-025-09538-w

[27] A. Abul Hasanaath, H. Luqman, R. Katib, and S. Anwar, “Fsbi: Deepfake detection with frequency enhanced self-blended images,” *Image and Vision Computing*, vol. 154, p. 105418, 01 2025. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S026288562500006X?via%3Dihub

References IX

[28] R. Sunil, P. Mer, A. Diwan, R. Mahadeva, and A. Sharma, “Exploring autonomous methods for deepfake detection: A detailed survey on techniques and evaluation,” *Heliyon*, vol. 11, no. 3, Feb 2025.

[29] M. Nasser, N. I. Arshad, A. Ali, H. Alhussian, F. Saeed, A. Da’u, and I. Nafea, “A systematic review of multimodal fake news detection on social media using deep learning models,” *Results in Engineering*, vol. 26, p. 104752, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590123025008291

[30] Y. Song, J. Kuang, and R. Xu, “Combating identity forgery in social media: Detecting manipulated profile videos with a motion-texture fusion framework,” *IEEE Access*, vol. 14, pp. 36 036–36 050, 2026.

[31] I. U. Haq and K. M. Malik, “Domain-adaptive representation learning for multimodal deepfake detection,” *IEEE Transactions on Computational Social Systems*, pp. 1–13, 2025.

References X

[32] S. M. Qureshi, A. Saeed, S. H. Almotiri, F. Ahmad, and M. A. Al Ghamdi, “Deepfake forensics: a survey of digital forensic methods for multimodal deepfake identification on social media,” *PeerJ Computer Science*, vol. 10, p. e2037, 2024.

Thank You

Any Questions?